

cal connection of a person's skin to the ground. The standard document completely describes the parameters for evaluation, acceptance and testing. Though the document describes the whole set of mechanical and electrical parameters over which a wrist strap needs to be evaluated and accepted, the most important parameter amongst all is the wrist strap continuity and resistance which should be one mega-ohm ± 20 per cent, for acceptance. The same applies to shoe straps. Figs 3 and 4 show the wrist strap and shoe strap, respectively.

ESD garments.

ESD garments such as apron, glove and shoe covers help in reducing the generation of static charge. The standard document ANSI/ESD STM2.1 defines the test methods for determining the electrical resistance from sleeve to sleeve and point to point of static control garment. The resistance range suggested by the document is 1×10^9 ohms to 1×10^{11} ohms, but most of the manufacturers do mention that the garments are tested as per ANSI/ESD STM2.1 standard, so do look for that. Fig. 5 shows an ESD-safe apron.

Work surface grounding. It is extremely important to ground the work surfaces, and the work surfaces that we will discuss here are:

Workstation surface. ESD mats should be used to cover the complete bench-top. The mat needs to be grounded and the best practice is to use metal grounding hardware snaps and ground cords connecting the work surface mat to the common-point ground. ANSI/ESD S4.1 standard provides test methods for evaluating and selecting work surface

TABLE VI
ESD Safety Solutions available from Different Manufacturers

ESD safety solutions	Types of solutions	Electronic Control Components and Devices	Static Systems Pvt Ltd	Sumitron Exports Pvt Ltd	Blue Sky System	Advance Tech Services Pvt Ltd
Personal grounding and isolation	Wrist straps	✓	✓	✓	✓	✓
	Footwear/shoe covers/shoe straps	✓	✓	✓	✓	✓
	Gloves	✓	✓	✓	✗	✓
	Apron/overcoat	✓	✓	✓	✓	✓
Antistatic work surfaces	Antistatic flooring mats	✓	✓	✓	✓	✗
	Antistatic flooring tiles	✓	✓	✓	✗	✗
	Antistatic table mats	✓	✓	✓	✓	✓
	Grounding cords	✓	✓	✓	✓	✓
ESD workstation	Table	✓	✓	✓	✓	✗
	Chairs	✓	✓	✓	✓	✗
	Stools	✓	✓	✓	✓	✗
	Cabinets	✓	✓	✓	✓	✗
	Brush	✓	✓	✓	✓	✓
	Ionisers	✓	✓	✓	✓	✓
ESD-safe packaging	Multilayer static shielding metallised bags	✓	✓	✓	✗	✗
	Black conductive bags	✓	✓	✓	✗	✗
	Pink ESD bags	✓	✓	✗	✗	✗
	Pink ESD bubble bags	✓	✓	✗	✗	✗
	ESD foam	✓	✓	✗	✗	✗
Testing instruments	ESD boxes	✓	✓	✓	✗	✗
	Wriststrap/footwear tester	✓	✓	✓	✓	✓
	Surface resistivity meter	✓	✓	✓	✓	✓
Miscellaneous	Electrostatic field meter	✓	✓	✓	✓	✓
	Bins	✓	✓	✓	✓	✗

materials, testing of new work surface installations and the testing of previously installed work surfaces. The guidelines for work surface in the document are mentioned below:

1. Resistance-to-groundable point 1×10^9 to 1×10^{10} ohms
2. Resistance from point to point ≥ 1 mega-ohm

These guidelines represent a range of resistance that has generally been proven to provide protection in the manufacturing environment. If the mat's resistance is too low, static transfers to the mat will be so fast that a spark is created. This spark is an ESD and will damage electronic devices.

If the mat's resistance is too high, static transfers so slowly that items placed on the mat will not lose their charge. When the item is removed from the mat, it will still have a static charge



Fig. 5: ESD-safe apron



Fig. 6: ESD mat

and be capable of discharging to other items. Fig. 6 shows an ESD mat.

Flooring. First decide on which type of flooring you will use: conductive (electrical resistance to ground range: 2.5×10^4 to 1.0×10^9) or dissipative (electrical resistance to ground range: 1.0×10^9 to 1.0×10^{10}).

Dissipative floors are lowest in cost but, if the floor is used as a primary ground for people and carts, a conductive floor should be chosen. ANSI/ESD S20.20 requires a system resistance of a person through the floor and to ground of less than 3.5×10^7 ohms (35 mega-ohms) when you use the floor as the primary ground. S20.20 allows static